

VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION–LANGUAGE MODELS

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ABSTRACT

Vision–Language Models (VLMs) are compelling candidates for zero-shot goal-conditioned value functions, yet their frozen pre-trained representations impose a fundamental ceiling on both generalization and temporal reasoning. We introduce VITA, a zero-shot value function learning framework that overcomes both limitations through meta-learned test-time adaptation. At inference, a lightweight adaptation module is updated by a single gradient step on a meta-learned self-supervised loss whose objective is, by construction, that each test-time update improve the value estimate. Because these updates are applied sequentially along a trajectory, VITA encodes trajectory history directly into the parameters of the adaptation module, thereby overcoming the temporal-reasoning limitations that constrain prior VLM-based estimators. To preclude shortcut learning, we further introduce a dissimilarity-based sampling strategy that, during training, selects the semantically most diverse segments of each trajectory. On real-world robotic manipulation, VITA generalizes from a single training environment to a broad range of out-of-distribution tasks, environments, and embodiments, consistently surpassing the state-of-the-art zero-shot estimator built on autoregressive VLMs. Moreover, VITA’s zero-shot value estimates furnish an effective dense reward for offline reinforcement learning: on the Meta-World MT10 benchmark, the resulting multi-task policies exceed those trained with the simulator’s hand-engineered fuzzy-logic dense rewards.

1 INTRODUCTION

Vision-Language Models (VLMs) achieve strong generalization across diverse tasks and domains by learning from large-scale, unstructured web data without human supervision (Radford et al., 2021; Alayrac et al., 2022). Generalist decision-making agents have not yet attained a comparable degree of transfer: despite substantial progress in learning generalist policies for robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024), prevailing methods remain bound to expert demonstrations (Ho & Ermon, 2016) and inherit their attendant cost and coverage limitations. World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) offer a compelling route around this dependence, allowing agents to learn within simulated experience across domains as varied as robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). Yet even as world models render increasingly realistic visual trajectories in diverse, open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), one question remains decisive: *how can agents learn effectively from video at scale without human supervision?*

One line of research learns policies directly from expert visual trajectories by inferring actions from visual transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a); the resulting latent actions can, in principle, be mapped to executable controls at deployment, though grounding them reliably remains difficult in practice. A complementary and increasingly influential line instead learns a universal *goal-conditioned value function* that serves, without task-specific retraining, as a zero-shot reward-shaping signal for reinforcement learning (RL) and as a supervision signal for imitation learning (Chen et al., 2021; Ma et al., 2022). Within this framework, goal-conditioned value estimation admits a natural formulation: from a visual observation and a natural-language task description, predict how far an agent has progressed toward

task completion (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025). We adopt this formulation and take it as the central estimation problem of the paper.

Prior work has repurposed pre-trained contrastive VLMs (Radford et al., 2021) for zero-shot reward shaping, exploiting the similarity between task descriptions and visual observations within a shared multimodal representation space to ground progress in semantic context (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024). Such frame-level similarity, however, is temporally blind: it cannot disambiguate visually similar states that recur at different stages of a task—folding versus unfolding a shirt being the canonical example. Autoregressive VLMs (Alayrac et al., 2022) instead inject temporal context by conditioning on the full visual trajectory in the prompt, but they inherit from the chronologically ordered corpora used in pre-training a systematic bias toward monotonically increasing predictions (Ma et al., 2024). The two architectures thus fail in complementary ways, yet share a common root cause: both commit to frozen pre-trained representations for zero-shot prediction (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023), and it is precisely this frozen commitment that caps their generalization and their temporal reasoning (Pătrăcean et al., 2023). Existing routes to more generalizable value functions—large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), and demonstration-based reward specification (Sontakke et al., 2023)—all buy generalization at the cost of substantial data, supervision, or task-specific adaptation. This motivates a different strategy: adapting the representation itself, online, to each test trajectory.

We introduce VITA, a zero-shot value function learning method that resolves both limitations at once by adapting a contrastive VLM at test time, and that surpasses the state-of-the-art zero-shot approach (Ma et al., 2024) on real-world robotic manipulation. Where prior methods purchase generalization through large-scale pre-training or expert demonstrations, VITA instead adapts online to the semantic and temporal context of each trajectory, requiring neither. At inference, a lightweight adaptation module is updated—at negligible cost—by a single gradient step on a meta-learned self-supervised loss (Sun et al., 2024) whose meta-objective is precisely that each such update improve value estimation (Finn et al., 2017). Applying these updates sequentially along a test trajectory encodes trajectory history directly into the module’s parameters (Sun et al., 2024), an implicit memory that supplies the temporal reasoning prior estimators lack. To preclude shortcut learning (Geirhos et al., 2020), we introduce a dissimilarity-based sampling strategy that steers the estimator toward semantic rather than positional cues, and we validate its effect empirically. We evaluate VITA along the three core axes of value-function competence—generalization under distribution shift, discrimination of expert from non-expert trajectories, and reward shaping for offline RL (Levine et al., 2020)—and find it effective on all three.

Our contributions are threefold.

- **Method.** We propose VITA, a meta-learned test-time adaptation method that endows contrastive VLMs with the generalization and temporal reasoning required for zero-shot value function estimation, while dispensing entirely with task-specific demonstrations and large-scale pre-training. Its temporal reasoning arises from an implicit memory built by sequential, per-timestep parameter updates, and a dissimilarity-based sampling strategy protects that memory from shortcut learning.
- **Generalization.** Trained in a single environment, VITA generalizes to diverse out-of-distribution tasks, environments, and embodiments in real-world robotic manipulation, outperforming the state-of-the-art zero-shot method (Ma et al., 2024) on every evaluated distribution shift.
- **Downstream utility.** Used as a zero-shot dense reward, VITA’s value estimates yield offline RL policies on the Meta-World multi-task benchmark (MT10) that surpass those trained with the simulator’s hand-engineered fuzzy-logic dense rewards (Yu et al., 2020)—despite VITA being trained only on real-world data.

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

For video trajectories, task progress estimation is naturally cast as goal-conditioned value function estimation in which the reward encodes task completion; we therefore formulate the learning of a goal-conditioned value function as the prediction, from vision–language representations, of the degree to which a task has been completed. Formally, the value function $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$ maps an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar estimate of progress toward goal completion, with $V(o_t; g) = 0$ denoting the start of the task and $V(o_t; g) = 1$ its completion. Following common practice, we identify task progress with temporal position in expert demonstrations, under the standard assumption that such trajectories advance monotonically toward goal completion (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Concretely, given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$ of length T , temporal progress is defined by the normalized timestep index $V^{\pi_E}(o_t; g) = \frac{t}{T}$, which thereby furnishes the supervisory target for learning V .

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) is a test-time adaptation paradigm that reframes inference as a self-supervised learning problem (Goodfellow et al., 2016), updating model parameters on each test instance without recourse to labels. First proposed for static image classification (Sun et al., 2020), TTT has since been extended to sequence modeling (Sun et al., 2024; Wang et al., 2023), where a parametric adaptation module f_{adapt} is updated at every timestep by a gradient step on a self-supervised loss. Whereas conventional sequence architectures retain temporal history in transient activation states—recurrent hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key–value pairs (Vaswani et al., 2017)—TTT instead writes that history into the parameters of f_{adapt} through its sequential inference-time updates. Those parameters consequently act as an implicit memory that accumulates temporal history while preserving temporal order. Sun et al. (2024) extend this construction by meta-learning the self-supervised task under the gradient-based meta-learning paradigm (Finn et al., 2017). In this setting (Sun et al., 2024), rather than being fixed a priori, the self-supervised loss is parameterized by learnable linear projections and optimized so that a test-time update reduces downstream prediction loss. We build on this formulation and apply it, for the first time, to goal-conditioned value function learning—capturing temporal context through the sequential updates and meta-learning the self-supervised task specifically to improve value estimation.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

Our goal-conditioned value function estimator is composed of three modules that operate in sequence: (1) a frozen multimodal encoder that extracts joint vision–language representations from the visual trajectory and its task description; (2) an adaptation module whose parameters are updated at test time under a self-supervised loss that has been meta-learned specifically to improve value estimation; and (3) a regression head that maps the adapted representation to a scalar value estimate. We describe each module in turn.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We adopt a frozen contrastive vision–language encoder, CLIP (Radford et al., 2021), to encode both visual observations and goal descriptions. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we form at each timestep t a joint multimodal representation by concatenating the two modality embeddings, yielding the sequence $(z_1, z_2, \dots, z_T)$ with $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, where ϕ_v and ϕ_g denote the visual and language encoders. Because CLIP’s contrastive pre-training aligns paired image–text inputs within a shared representation space (Radford et al., 2021), visual observations that are semantically nearer to goal completion are embedded closer to the goal description—a property that renders these representations a well-suited substrate for progress estimation.

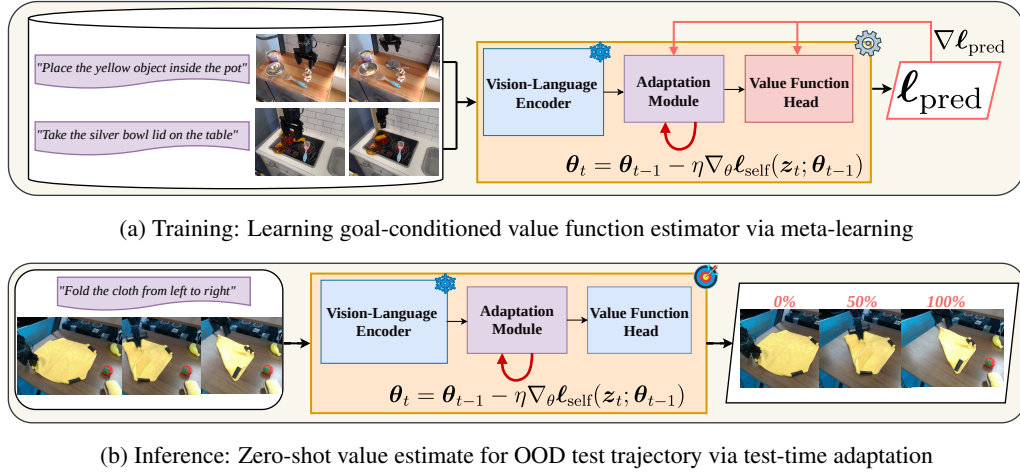


Figure 1: **Overview of VITA.** During training (top), VITA meta-learns a goal-conditioned value function estimator so that a subsequent self-supervised update improves value prediction; at inference (bottom), the same estimator adapts online to each out-of-distribution test trajectory, yielding zero-shot generalization via test-time adaptation.

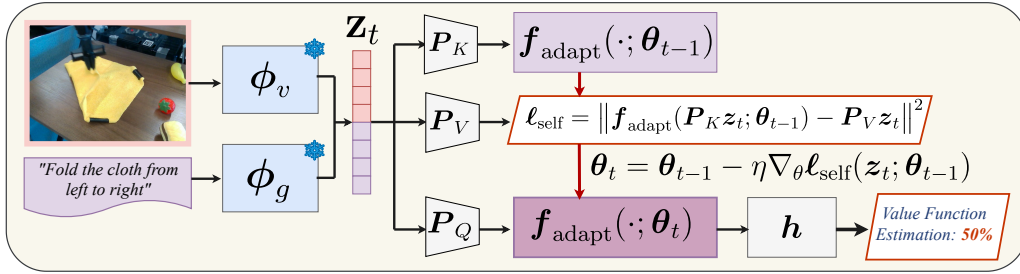


Figure 2: **Test-time Adaptation.** At each inference timestep t , the adaptation module f_{adapt} is updated by a single gradient step on the meta-learned self-supervised loss ℓ_{self} ; iterating this update along the trajectory accumulates temporal history in the module’s parameters, which thereby serve as an implicit memory.

3.1.2 TEST-TIME ADAPTATION

To align the multimodal representations with both the semantic and the temporal context of a given trajectory, we introduce an adaptation module f_{adapt} that follows the test-time training paradigm of Section 2.2. At inference, the parameters of f_{adapt} are updated online by a gradient step on a meta-learned self-supervised loss ℓ_{self} , which takes the form of a reconstruction objective mediated by two learnable linear projections, P_K and P_V , that are meta-learned during training so that adaptation improves value estimation. Concretely, with $P_K \in \mathbb{R}^{d' \times 2d}$ producing a projected input view and $P_V \in \mathbb{R}^{d' \times 2d}$ producing the reconstruction target, the self-supervised loss is defined as:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} before test-time adaptation. At each timestep t , the parameters are updated by

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . Because each update is conditioned on the parameters produced by all preceding updates, the sequential application of Eq. 2 causes f_{adapt} to retain information from past visual observations within its parameters, thereby capturing temporal context. In Section 4.6.2 we present empirical evidence that these incremental, per-timestep parameter updates yield substantially more accurate value estimation than a single update aggregated at the trajectory level.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

Once the adaptation step has been applied, a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps the input z_t into the adaptation space $\mathbb{R}^{d'}$; the projected representation is then transformed by the adaptation module f_{adapt} and, finally, by the regression head h , which emits the predicted value of the goal-conditioned value function:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), mirroring the value-function estimator architectures standard in deep reinforcement learning (Espeholt et al., 2018). It is trained on the normalized progress labels y_t from expert demonstrations defined in Section 2.1, under a supervised prediction loss ℓ_{pred} given by the mean squared error between the predicted values $V(z_t; g)$ and the targets y_t . This division of labor is deliberate: at inference the head h stays frozen, while only the adaptation module f_{adapt} is updated online through the self-supervised objective—so that the estimator generalizes zero-shot to novel trajectories without any task-specific demonstration.

3.2 TRAINING PROCEDURE

We train the value function estimator by gradient-based meta-learning, shaping the self-supervised loss ℓ_{self} so that the resulting test-time adaptation reduces the supervised prediction loss ℓ_{pred} (Sun et al., 2024). Training operates over diverse sub-trajectories selected by dissimilarity-based sampling, a mechanism that suppresses shortcut learning and directs the estimator toward semantic and temporal cues. The two components are described below.

3.2.1 META-LEARNING

During training, the adaptation module f_{adapt} is updated online at each timestep under the self-supervised loss ℓ_{self} , exactly as it will be at inference. Following the gradient-based meta-learning paradigm (Finn et al., 2017), we then backpropagate through these adaptation updates with respect to the supervised prediction loss ℓ_{pred} , jointly optimizing the adaptation initialization θ_0 , the linear projections P_K, P_V, P_Q , and the regression head h . The total training objective is the supervised prediction loss ℓ_{pred} augmented by the self-supervised loss ℓ_{self} , weighted by a scalar λ . Crucially, the learned initialization θ_0 is the very point from which adaptation proceeds at inference; meta-learning it in this way teaches the estimator not merely to predict progress, but to *adapt* its internal representations at test time so as to sharpen that prediction.

3.2.2 DISSIMILARITY-BASED SAMPLING

Expert visual trajectories are dominated by highly redundant consecutive frames, a property long noted in video action recognition (Wang et al., 2018). This redundancy is not benign: it invites the value function estimator to exploit shortcut cues (Geirhos et al., 2020; Lee et al., 2021b), overfitting to the frequently recurring late-stage visual patterns rather than to genuine task semantics. To counter this failure mode, we propose a dissimilarity-based sampling strategy that assembles each mini-batch from the most visually diverse sub-trajectories within a trajectory, raising intra-batch variance and thereby acting as a form of importance sampling that up-weights semantic diversity.

Formally, from a sequence of multimodal representations $\{z_1, \dots, z_T\}$ we extract fixed-length sub-trajectories with a sliding window of size w_{tr} and stride s , obtaining a candidate set $\mathcal{W}$. The maximally diverse subset of size k is the one that maximizes total pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i is a sub-trajectory of length w_{tr} . Solving Eq. 4 exactly is combinatorially intractable, as it requires enumerating all $\binom{|\mathcal{W}|}{k}$ subsets. We therefore approximate the objective with an efficient scoring-based heuristic that assigns each window $w \in \mathcal{W}$ a diversity score equal to its aggregate dissimilarity to every other window:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

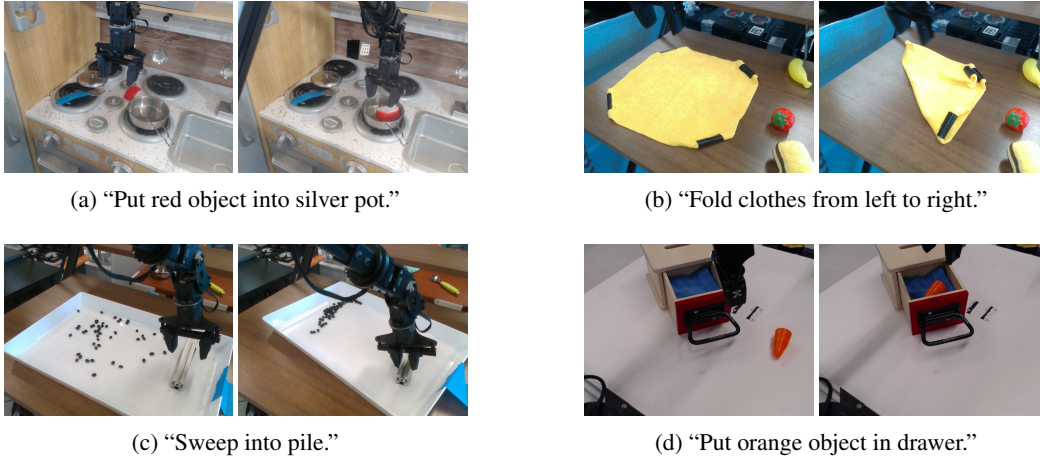


Figure 3: Representative visual trajectories, each paired with its task description, spanning the distribution shifts on which VITA is evaluated: (a) in-distribution; (b, c) environment shift; (d) joint embodiment and environment shift.

Computing the score s for every window and retaining the k highest-scoring ones thus yields a diverse subset in polynomial rather than exponential time. The resulting overhead is negligible relative to model training, as our complexity analysis in Appendix G establishes. Our ablation study (Section 4.6.1) confirms the practical payoff: relative to full-trajectory sampling, dissimilarity-based sampling markedly sharpens the estimator’s ability to distinguish expert from non-expert visual trajectories.

4 EXPERIMENTS

Our evaluation is organized around the three core capabilities that a goal-conditioned value function must exhibit, and it establishes VITA’s effectiveness on each: (i) generalization of value estimation for real-world robotic manipulation under distribution shifts in task, environment, and embodiment (Table 1); (ii) discrimination between expert and non-expert visual trajectories in real-world robotic manipulation (Table 2a); (iii) zero-shot reward shaping for offline RL on Meta-World MT10, a simulated multi-task robot-learning benchmark (Table 2b). Together these results substantiate the paper’s central claim: meta-learned test-time adaptation yields a single value function estimator that generalizes, discriminates, and shapes rewards, without task-specific retraining.

4.1 TRAINING SETUP

We train VITA on expert visual trajectories paired with natural-language task descriptions drawn from the BridgeData V2 dataset (Walke et al., 2023), which spans a broad range of manipulation tasks, environments, and robot embodiments. Deliberately, we restrict training to a narrow slice of this diversity: a curated subset of 2,986 expert demonstrations of pick-and-place manipulation only—excluding folding, sweeping, and stacking—collected with a single robot embodiment (WidowX 250) across four configurations of the ToyKitchen environment. This constrained training regime makes the generalization results that follow a stringent test rather than an in-distribution one. A further 287 expert demonstrations are held out as the in-distribution test set, denoted tk_pnp , with training examples shown in Appendix A. We adopt OpenCLIP ViT-B/32 as the frozen backbone for encoding video frames and task descriptions. The training objective combines ℓ_{pred} with ℓ_{self} at weight $\lambda_{\text{self}} = 0.5$, and dissimilarity-based sampling is configured with window size $w_{\text{tr}} = 8$, sub-trajectory count $k = 8$, and stride $s = 1$. At test time we apply a single gradient step ($t_{\text{ep}} = 1$) under λ_{self} with learning rate $\eta = 0.1$. Complete hyperparameters and sweeps are reported in Appendix D.

4.2 BASELINES

We benchmark VITA against value functions spanning both contrastive and autoregressive VLM paradigms, so that its gains can be attributed to temporal modeling rather than to any single architecture. VLM-CL derives zero-shot value estimates from the cosine similarity between the CLIP representations of each frame and the task description (Baumli et al., 2023). VLM-RM regularizes CLIP embeddings by projecting features onto the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT trains a supervised regression head on frozen CLIP multimodal representations, and CLIP-GRU trains a Gated Recurrent Unit (GRU) (Cho et al., 2014) on the same representations, providing a recurrent temporal baseline. GVL (Ma et al., 2024) is the state-of-the-art zero-shot value function built on autoregressive VLMs; we evaluate two settings—GVL-0S, the zero-shot regime in which the VLM is prompted with only the test trajectory and task description, and GVL-1S, a one-shot in-context regime that additionally supplies a full shuffled training-distribution trajectory with its progress labels as an exemplar. Following GVL, we use Gemini 1.5 Pro (gemini-1.5 Pro-latest) (Gemini Team et al., 2024) with its proposed prompt template. We further probed two open alternatives under the same protocol: Qwen-VL 2.5 (Yang et al., 2024) failed to overcome the temporal bias, and GPT-4o (Hurst et al., 2024) declined to produce value estimates. Additional implementation details appear in Appendix D.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

Table 1: Value Order Correlation (VOC) for value function estimation under distribution shift; higher is better, with the best per row in **bold** and the second-best underlined. Across the ten evaluation datasets, VITA attains the best score on six and the second-best on the remainder, and it surpasses the state-of-the-art zero-shot estimators GVL-0S and GVL-1S on every dataset. Shift types: **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = both shifts jointly.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	<u>0.658</u>
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

We assess how well progress estimators generalize across novel environments, tasks, and robot embodiments using subsets of BridgeData V2 curated to introduce controlled variation along each axis. Environment shifts alter scene layout or background: `lm_pnp` is a pick-and-place task in front of a laundry machine, `td_fold`, `ft_fold`, and `rd_fold` involve cloth folding on distinct surfaces, and `ms_sweep` is a long-horizon sweeping task in a confined tray. Embodiment shifts employ the DeepThought robot, which departs from the training embodiment (WidowX 250) in both morphology and camera perspective: `dt_tk_pnp` (pick-and-place) and `dt_tk_stack` (stacking) preserve the in-distribution environment under the new embodiment, whereas `dt_ft_stack` (stacking) and `dt_rd_pnp` (drawer pick-and-place) compound embodiment and environment shifts. Each dataset comprises 200 expert trajectories with task descriptions, save for `ms_sweep`, which contains 100 owing to its limited size; Appendix B describes all evaluation datasets in full. We quantify performance with the *Value Order Correlation* (VOC) metric (Ma et al., 2024), which measures the alignment between predicted progress and the chronological order of frames in a trajectory. Formally, letting $p_1, \dots, p_T$ denote the predicted progress values over a trajectory of T frames and $r_t = t$ the temporal index, VOC is the Spearman rank correlation ρ_s between the two.

Table 1 makes the central comparison legible. Both VITA and CLIP-GRU—the two temporal-modeling methods—outperform GVL-0S and GVL-1S on every dataset, a uniform margin that

exposes the limitations of pre-trained autoregressive VLMs and confirms that preserving temporal order is essential to progress estimation. Between the two temporal methods, VITA is the stronger: it attains the higher VOC on six of the ten datasets, and where CLIP-GRU leads, the two remain close. Both, in turn, decisively outperform the temporally blind CLIP-based baselines (CLIP-FT, VLM-CL, VLM-RM), isolating the encoding of temporal history as the decisive factor. CLIP-FT matches GVL in-distribution but collapses under distribution shift, while VLM-CL and VLM-RM perform poorly throughout, consistent with their absence of temporal modeling. For GVL, the one-shot in-context example is unreliable: relative to GVL-0S, GVL-1S helps on some datasets, is neutral on others, and degrades performance on still others. Tellingly, both GVL variants score well on folding tasks (`td_fold`, `ft_fold`, `rd_fold`) yet falter on stacking (`dt_tk_stack`, `dt_ft_stack`) and pick-and-place (`dt_tk_pnp`, `lm_pnp`), a pattern that points to a bias in the underlying autoregressive VLM toward folding-like manipulations. VITA, by contrast, sustains uniformly strong performance across every task type and every distribution shift—evidence of markedly stronger generalization than in-context learning affords.

The long-horizon task `ms_sweep` is the hardest setting—every method except CLIP-FT degrades on it—yet VITA still records the highest VOC (0.490), edging out CLIP-GRU (0.434). This margin is instructive: for long-horizon manipulation, encoding temporal history through sequential test-time adaptation proves more robust than compressing it into recurrent hidden states. VITA generalizes equally well under embodiment shift. On `dt_tk_pnp`, which transfers to a new robot while retaining the training environment and task, both VITA and CLIP-GRU *exceed* their own in-distribution VOC, demonstrating that a learned temporal-context representation transfers cleanly across embodiments. GVL-1S, by contrast, performs poorly on both `dt_tk_pnp` and `dt_tk_stack`, indicating that its few-shot in-context mechanism does not generalize under the embodiment shifts in our evaluation.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Generalization aside, a dependable value function must also be discriminative: it must assign lower progress to suboptimal behavior than to expert behavior. We test this property by contrasting VITA’s predictions on expert versus scripted (non-expert) visual trajectories drawn from the same in-distribution setting of BridgeData V2. The scripted trajectories are produced in the ToyKitchen environment by a heavily randomized controller (Walke et al., 2023), under the identical embodiment and environmental configuration as the expert demonstrations, and span pick-and-place tasks over object categories that may overlap with training—general objects (`sc_pnp_obj`), rigid items (`sc_pnp_robj`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and a cloth (`sc_pnp_cloth`). Although matched to the training distribution in task, embodiment, and environment, their execution departs from the smooth, monotonic progress characteristic of expert demonstrations; Appendix C shows examples. Because the randomized controller does not target calibrated levels of suboptimality, these datasets are not designed to encode quantifiable degrees of it; nonetheless, an effective estimator should, in the same setting, assign systematically lower VOC to suboptimal than to expert trajectories. We capture this with *BinVOC*, the binary criterion $1[\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, equal to 1 when the expert trajectory attains higher VOC than its suboptimal counterpart and 0 otherwise.

We report discriminative performance as BINVOC averaged over all scripted datasets, with expert performance (VOC_{exp}) measured on the in-distribution test set `tk_pnp`, whose pick-and-place tasks parallel those in the scripted evaluation. As Table 2a shows, VITA achieves perfect discrimination ($\text{BinVOC} = 1.00$), matching GVL-0S and GVL-1S in consistently ranking every expert trajectory above its suboptimal counterpart. Notably, VITA surpasses CLIP-GRU (0.80) here, suggesting that temporal memory held in recurrent hidden states is more susceptible to overfitting temporal shortcuts than the implicit memory formed by sequential test-time updates. The temporally blind CLIP baselines VLM-CL (0.40) and VLM-RM (0.00) fare worst, again consistent with their lack of temporal modeling, while CLIP-FT is comparatively reliable (0.80) but fails on one task, exposing a sensitivity to object distribution shift that VITA does not exhibit.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

To establish downstream utility, we evaluate our estimator as a zero-shot reward source for offline RL on the Meta-World MT10 benchmark (Yu et al., 2020), ten diverse robotic manipulation tasks

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-0S	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: (a) Discrimination of expert from scripted robot demonstrations, as average BinVOC over the 5 in-distribution scripted datasets (higher is better); VITA reaches the perfect score of 1.00, matching GVL and exceeding all other learned baselines. (b) Offline RL performance on Meta-World MT10, as interquartile mean (IQM) return with 95% stratified bootstrap CIs; VITA achieves the highest IQM (0.815) and surpasses the simulator’s fuzzy-logic dense reward (META-WL, 0.779). Best result per panel in **bold**.

spanning pick-and-place, door opening, and pushing. The transfer here is severe and deliberate: VITA is trained solely on real-world robotic data and applied, without any adaptation of its training, to this simulated domain. For each MT10 task we generate 20 expert visual demonstrations from Meta-World’s expert policies, then use the goal-conditioned value estimator to assign a dense reward to each visual observation, and train policies with Implicit Q-Learning (IQL) (Kostrikov et al., 2021) in the offline RL regime. Demonstration generation and policy training are repeated over 10 random seeds; each resulting multi-task policy is evaluated on 20 rollouts per task, and we compute the average return per (task, seed) pair. We report the interquartile mean (IQM) across all pairs with 95% stratified bootstrap confidence intervals, following the protocol of Agarwal et al. (2021); full training and evaluation details are given in Appendix E.

Table 2b reports MT10 performance under this protocol. VITA attains the highest IQM return, 0.815 (95% CI [0.785, 0.838]), outperforming every CLIP-based baseline. A like-for-like comparison with GVL is infeasible at this scale, since GVL relies on the proprietary Gemini-1.5 VLM, whose per-query inference cost is prohibitive in the RL loop, and the open-source alternatives we tried did not yield reliable progress estimates in preliminary tests. The decisive comparison is instead against the simulator’s own hand-engineered fuzzy-logic dense reward (META-WL, 0.779): despite being strong, META-WL is surpassed by VITA, and the confidence intervals underscore the gap. That a value estimator trained exclusively on real-world data can outperform a reward function purpose-built for the simulator demonstrates that VITA’s zero-shot value estimates transfer effectively into simulated reward shaping.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We isolate the contribution of dissimilarity-based sampling by holding the model fixed and varying only the training-time sub-trajectory sampling strategy, measuring the effect on VITA’s ability to separate expert from non-expert demonstrations. Full-trajectory sampling (Sun et al., 2024), which computes the adaptation loss over all overlapping sub-trajectories at stride $s = 1$, overfits in our setup to global temporal shortcuts. Random sampling injects diversity by drawing sub-trajectories uniformly, but its diversity is undirected and semantically uninformed. Dissimilarity-based sampling instead deliberately selects the sub-trajectories that maximize pairwise dissimilarity, and this targeted semantic diversity is what yields the strongest discrimination: our method outperforms both full-trajectory and random sampling at differentiating expert from non-expert visual trajectories. Appendix F.1 provides the full analysis.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We next dissect *how* temporal context should be incorporated at test time, comparing adaptation strategies that differ only in their memory mechanism. VITA adapts step-by-step without reset

(implicit memory), incrementally updating parameters from θ_{t-1} so that trajectory history is encoded directly into the parameters, preserving both temporal order and accumulated context. We contrast it with three alternatives that each ablate one facet of this mechanism: (i) *trajectory-level adaptation (TTT-TR)*, a single gradient update over the full trajectory—a batched update that averages the loss across all frames and thereby discards temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets the module to θ_0 at every step and adapts only to the current observation, carrying no temporal information forward; (iii) *explicit memory (TTT-EX)*, which likewise resets at every step but adapts over a local window of recent frames matching the meta-training window ($w_{tr} = 8$). VITA consistently outperforms all three across every dataset, establishing that the incremental accumulation of temporal context—rather than temporal information per se—is what drives accurate value estimation. Appendix F.2 reports the full analysis.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Contrastive VLMS have been deployed as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024), relying on frame-level similarity scores that carry no temporal modeling. A related family fine-tunes CLIP on domain-specific video (Fan et al., 2022; Jiang et al., 2024) to produce dense rewards, but the required supervision curtails zero-shot applicability. RoboCLIP (Sontakke et al., 2023) permits one-shot reward specification without fine-tuning, yet yields sparse rewards and still depends on demonstrations. Large-scale multimodal pre-training approaches (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) learn language-conditioned value functions by leaning on scale for generalization, and ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data with video rewinding, coupled with a handful of in-domain demonstrations. VITA departs from all of these by adapting a pre-trained contrastive vision–language encoder *at inference*, capturing temporal and semantic context through test-time adaptation and requiring neither large-scale multimodal pre-training nor additional domain-specific demonstrations to generalize. Because it is agnostic to the underlying encoder, VITA applies to any pre-trained vision–language representation of a universal value function, enhancing its generalization and temporal reasoning through meta-learned test-time adaptation. Autoregressive VLMS (Alayrac et al., 2022) exploit in-context learning over trajectories and have been applied to success detection (Du et al., 2023) and goal-conditioned value functions (Ma et al., 2024). GVL (Ma et al., 2024) counters the monotonic bias induced by ordered pre-training data by shuffling frames at inference—but at the cost of discarding temporal order, the very signal needed for temporal reasoning and for disambiguating visually similar yet temporally distinct states. VITA instead *preserves* chronological order while capturing temporal context through test-time adaptation, obtaining the benefits of temporal modeling without the shortcut GVL must accept.

5.2 TEST-TIME ADAPTATION

Test-time adaptation offers a parameter-efficient alternative to full fine-tuning for adapting VLMS. Prompt-based methods optimize model parameters at test time from task context, though the mechanisms by which they succeed remain opaque (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) argue that CLIP representations are amenable to test-time adaptation, motivated by the finding that CLIP encodes human-interpretable concepts identified through mechanistic interpretability (Bricken et al., 2023); CLIP-based similarity has also served as a reward whose parameters are optimized at test time via reinforcement learning (Zhao et al., 2024). Test-time training (Sun et al., 2020), in which an adaptation module is updated at each timestep by a self-supervised loss, is a further instance. Unlike meta-reinforcement learning (Finn et al., 2017; Bauer et al., 2023), which demands task-specific adaptation episodes at training time, test-time training adapts directly and online during inference without task labels—the property VITA builds upon.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

Our experiments establish that meta-learned test-time adaptation equips value functions for robotic manipulation to generalize across distribution shifts in task, environment, and embodiment. Beyond generalization, VITA discriminates expert from non-expert visual trajectories and performs zero-shot reward shaping, opening downstream applications in RL and imitation learning. The common source of these capabilities is the sequential, per-timestep update rule: by writing trajectory history into its parameters, VITA captures temporal and semantic context more effectively than either CLIP-based baselines or in-context learning with autoregressive VLMs. That VITA outperforms CLIP-GRU on both expert–non-expert discrimination and offline RL indicates that, for zero-shot value estimation, implicit memory accrued through sequential test-time updates generalizes more reliably than temporal history compressed into recurrent hidden states. Pre-trained autoregressive VLMs such as GVL, for all their generalization, are not trained for progress estimation and therefore lack the temporal reasoning that task requires. Finally, the per-timestep adaptation carries negligible inference-time overhead—only a lightweight module is updated—so VITA’s advantages come without compromising real-time applicability.

6.2 LIMITATIONS AND FUTURE WORK

While test-time adaptation improves the generalization of zero-shot progress estimation across the unseen tasks and environments in our experiments, settings with high execution variability or extended duration remain challenging. And although the lightweight adaptation module keeps the adaptation overhead negligible, updating a value estimator at every timestep may be unsafe in deployment, which tempers applicability in the most demanding real-time scenarios. These considerations delineate the natural next steps. We plan to investigate alternative test-time adaptation schemes for vision–language models, particularly for training agents within world models, and to extend VITA to real-time closed-loop control and more complex RL environments. Because VITA is agnostic to the underlying vision–language encoder, it stands to enhance the generalization and temporal reasoning of any domain-specific pre-trained multimodal representation through meta-learned test-time adaptation. Finally, although we establish empirically that dissimilarity-based sampling mitigates shortcut learning, a theoretical account of how diversity-based sampling shapes shortcut learning remains open; this lies beyond the scope of our core contribution—a value function estimator that meta-learns a test-time adaptation mechanism to improve zero-shot performance and temporal reasoning—but constitutes a promising direction.

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A TRAINING DATASET

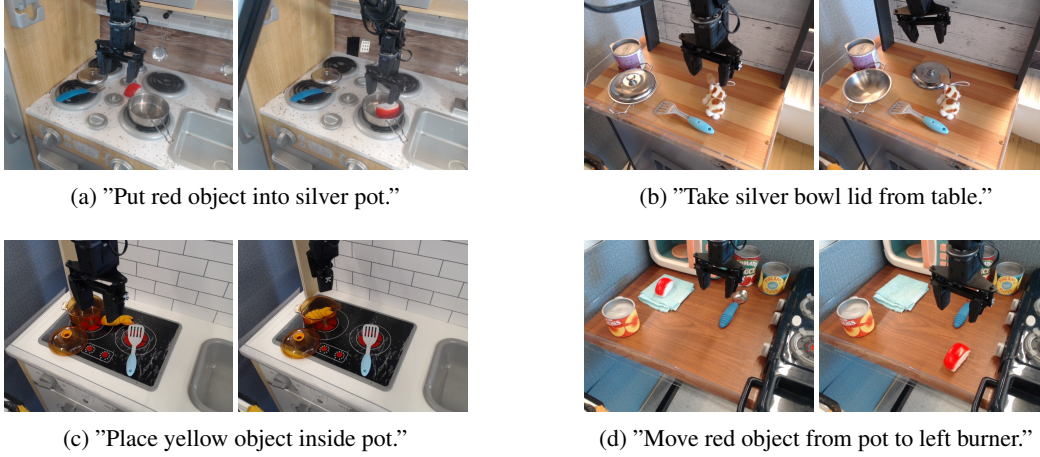


Figure 4: Representative training demonstrations: each subfigure pairs the start and end frames of an expert demonstration with its natural-language task description. All demonstrations are pick-and-place tasks collected with the WidowX 250 embodiment across four distinct ToyKitchen configurations.

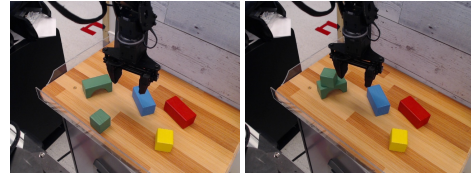
B EVALUATION DATASETS

Table 3: Evaluation datasets, characterized by task type, environment, and embodiment. Together they span the four distribution-shift regimes—in-distribution, environment shift, embodiment shift, and joint shift—used throughout the evaluation.

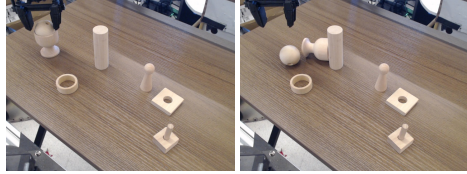
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

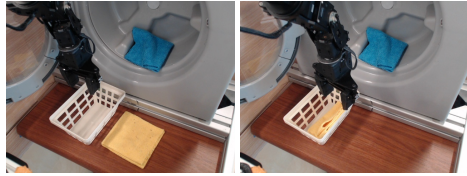


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

Figure 5: Evaluation trajectories under embodiment shift: each subfigure pairs the start and end frames with the task description. The top row isolates embodiment shift—the same ToyKitchen environment operated by a different robot (DeepThought)—while the bottom row compounds it with environment shift by introducing new scenes.



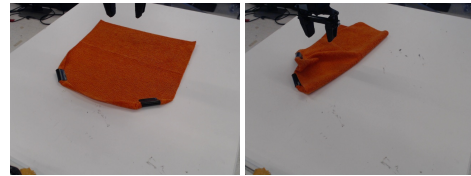
(a) lm_pnp: "Place blue cloth inside washer."



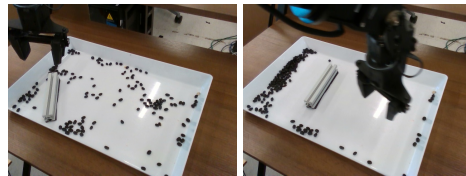
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Evaluation trajectories under environment shift: each subfigure pairs the start and end frames with the task description. All tasks retain the training embodiment (WidowX 250) while varying the scene, isolating the effect of visually and structurally distinct environments.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Scripted (non-expert) evaluation trajectories: each subfigure pairs the start and end frames of a scripted trajectory, collected in the ToyKitchen environment with the WidowX 250 robot, with its task description. All are pick-and-place behaviors spanning distinct object categories; task descriptions are generic, as per-trajectory annotations are not publicly available.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

We adopt the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset as a frozen backbone, encoding each visual observation with the CLIP vision encoder and each task description with the text encoder. We form the joint multimodal representation by concatenation rather than element-wise product, a choice made on the basis of superior validation performance. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activation and projection dimension $d' = 64$. Optimization uses AdamW with learning rate 1×10^{-4} , weight decay 1×10^{-4} , and a cosine schedule with 10% warmup, at batch size 32; all trajectories are padded to a maximum length of 120 frames, matching the longest training sequence. The self-supervised weighting coefficient λ_{self} is fixed at 0.5, again selected on validation performance. We train for 5 epochs, since validation VOC plateaus early and further training yields no improvement. Unless stated otherwise, dissimilarity-based sampling uses stride $s = 1$, sub-trajectory length $w_{\text{tr}} = 8$, and $k = 8$ selected windows. All experiments were run on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference we adapt only the temporal adaptation module f_{adapt} , retaining the training-time projection dimension ($d' = 64$), and perform a single gradient step ($t_{\text{ep}} = 1$) at learning rate $\eta = 0.1$. This configuration was chosen from a sweep over learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation steps $\{1, 5, 10\}$ by validation performance (Table 4). Two regimes prove essentially

Table 4: Sensitivity of validation VOC to the test-time learning rate η and the number of adaptation steps t_{ep} . The best VOC in each column is shown in **bold**; performance peaks along a diagonal trading off step size against step count, and our chosen configuration ($\eta = 0.1$, $t_{\text{ep}} = 1$) attains the overall best value (0.783).

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

equivalent: a single step at the larger rate ($\eta = 0.1$, $t_{\text{ep}} = 1$) and ten steps at the smaller rate ($\eta = 0.01$, $t_{\text{ep}} = 10$). We adopt the former, since it matches—indeed slightly exceeds—the latter’s validation VOC while requiring an order of magnitude fewer adaptation steps, which directly minimizes inference-time overhead.

D.3 BASELINES

VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rate $\eta = 0.1$, all fixed via the hyperparameter selection described above. The baselines are constructed to hold architecture fixed and isolate the mechanism under study. CLIP-FT mirrors VITA’s architecture but strips away the adaptation module and self-supervised loss: a frozen CLIP encoder feeds a linear projection and a two-layer MLP that predicts task progress under standard supervised regression, with neither meta-learning nor test-time adaptation. To ensure this ablation is not merely underparameterized, we grant it additional capacity—an $8\times$ larger projection matrix and $10\times$ more training steps than VITA (Lin et al., 2022). For VLM-RM we use a baseline prompt describing the environment. CLIP-GRU likewise retains VITA’s frozen CLIP encoder and regression head h but substitutes a single-layer Gated Recurrent Unit of hidden size 64 for the adaptation module f_{adapt} , encoding temporal history explicitly through recurrent hidden states and passing the final hidden state to h ; this makes CLIP-GRU the controlled counterpart that differs from VITA only in its memory mechanism. For both GVL-0S and GVL-1S we use the latest Gemini 1.5 Pro, gemini-1.5-pro-latest (Gemini Team et al., 2024), updating the earlier gemini-1.5-pro release of the original GVL implementation. We additionally evaluated GVL with the open-source autoregressive VLM Qwen-VL 2.5 (Yang et al., 2024), which failed to overcome temporal bias despite frame shuffling, and GPT-4o (Hurst et al., 2024), which consistently declined to produce scalar progress estimates in our setup. All GVL evaluations followed the prompt template of the original work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

We train with Implicit Q-Learning (IQL) under a single configuration held fixed across all MT10 tasks, so that any performance difference is attributable to the reward source rather than to per-task tuning. Expectile regression uses $\tau = 0.7$, with advantage-weighting temperature 3.0 and clipping threshold 100.0; the actor and critic use learning rates 1×10^{-4} and 3×10^{-4} respectively, at batch size 256. Policies are trained for 100,000 gradient steps and evaluated every 10,000 steps on 20 rollouts per task (horizon 150). The implementation follows d3rlpy with configuration expectile=0.7, weight_temp=3.0, max_weight=100.0, actor_lr=1e-4, critic_lr=3e-4, batch_size=256.

E.2 EVALUATION PROTOCOL

Following Agarwal et al. (2021), we report the interquartile mean (IQM) of returns across all task–seed pairs. The IQM—the mean of the middle 50% of scores—is far less sensitive to outliers

than either the mean or the median, and is therefore the appropriate summary statistic for the multi-task, multi-seed regime evaluated here. We accompany it with 95% confidence intervals estimated by stratified bootstrap over 10,000 resamples, stratified by task. This protocol is applied identically to every method in the Meta-World MT10 experiments, ensuring that the reported comparisons are statistically commensurable.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5: Per-dataset Disc@VOC across the scripted datasets under different training-time sampling strategies for test-time adaptation; a checkmark marks successful expert-versus-suboptimal discrimination. Sampling methods—**FT** (full-trajectory), **Rand** (random), and **Diss** (dissimilarity-based)—are parameterized by (w_{tr}, k) . Dissimilarity-based sampling at (8, 8) is the only configuration to discriminate perfectly on all five datasets (avg. 1.0), whereas full-trajectory sampling discriminates on only one (avg. 0.20).

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

We validate the proposed dissimilarity-based sampling by varying only the training-time sub-trajectory sampling strategy and measuring its effect on the model’s ability to discriminate expert from sub-optimal trajectories. In Table 5 we report Disc@VOC over the 5 scripted datasets, which records whether a model assigns higher average VOC to expert trajectories (`tk_pnp`) than to each evaluation dataset. Three strategies are compared: (1) full-trajectory sampling, which adapts over all overlapping sub-trajectories of length w_{tr} at stride s , following prior work (Sun et al., 2024); (2) random sampling, which draws k sub-trajectories of length w_{tr} uniformly per video; and (3) our dissimilarity-based sampling, which builds a candidate set of length- w_{tr} sub-trajectories at stride s and selects the k that maximize pairwise dissimilarity in representation space.

The results are unambiguous. Dissimilarity-based sampling at $w_{tr} = 8$, $k = 8$ attains perfect discrimination across all five datasets; reducing to $k = 4$ windows of the same length preserves this perfect score, whereas shortening the sub-trajectory to $w_{tr} = 4$ degrades it—implicating window length, not merely count, as the operative factor. The full-trajectory baseline is weakest (avg. 0.20), confirming that adapting over all consecutive windows invites overfitting to global temporal cues, and random sampling, though it improves on full-trajectory (avg. 0.80), remains less reliable than dissimilarity-based selection. Taken together, these findings show that both sub-trajectory diversity and adequate semantic locality are necessary for discriminative test-time adaptation.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We evaluate four adaptation strategies, each ablating a distinct facet of temporal memory: (1) TTT-TR, a single offline gradient update over the full trajectory; by averaging the loss across all frames in one batched update, it discards temporal order entirely; (2) TTT-RS, which resets the adaptation module at every step and adapts to the current observation alone (memoryless), carrying no temporal information forward; (3) TTT-EX, which likewise resets at every step but adapts over a local window of recent frames matching the meta-training window ($w_{tr} = 8$); averaging gradients over this window discards the temporal ordering within it; (4) VITA (OURS), which updates the module incrementally at every timestep (implicit memory), forming its memory through the sequential accumulation of temporal history in the adapter parameters. All variants were subjected to the identical test-time hyperparameter sweep described in Appendix D. VITA, TTT-RS, and TTT-TR are optimal at

Table 6: Validation VOC for progress estimation under distribution shift, comparing the three ablated adaptation strategies against VITA (best per row in **bold**). VITA outperforms all three on every dataset by a wide margin, isolating sequential per-timestep updates as the source of its temporal reasoning. Shift types: ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = joint Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

learning rate 0.1 with a single adaptation step, while TTT-EX peaks at learning rate 1.0 with one step.

Table 6 reveals a sharp dichotomy. The three ablated strategies—TTT-TR, TTT-RS, and TTT-EX—cluster at comparably low VOC, showing that a batched update, whether over a short explicit window (TTT-EX) or over the full trajectory (TTT-TR), buys essentially nothing over the memoryless baseline (TTT-RS). VITA, by contrast, outperforms all three on every dataset by a large and consistent margin, establishing sequential per-timestep updates as the mechanism responsible for effective value estimation in real-world robotic manipulation. Two structural arguments explain why this mechanism is not merely preferable but necessary. First, within implicit memory, adapting over a local window would double-count the most recent observations—once explicitly through the window and once implicitly through the accumulated parameters. Second, updating less frequently than every timestep would omit intermediate frames and thereby lose information. The implicit-memory test-time training paradigm consequently *requires* sequential, per-timestep updates without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ denote the candidate set of size $N_c = |\mathcal{W}|$. To approximate diverse-subset selection without the exponential cost of enumerating all $\binom{N_c}{b}$ subsets in Eq. 4, we assign each candidate window a diversity score equal to the sum of its pairwise distances to all others—equivalently, the row sum of the distance matrix. With $N_c = T - w_{tr} + 1$ candidate windows, each represented by a flattened feature vector of dimension $m = w_{tr} \cdot d$ (where d is the dimension of each multimodal representation z_t), the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ is computed on the GPU via batched `cdist` at a cost of $\mathcal{O}(N_c^2 \cdot w_{tr} \cdot d)$ operations—polynomial in N_c rather than exponential.

In our experiments the training trajectories are short (mean length 69, maximum 120). With batch size 32, feature dimension $d = 1024$ for the CLIP-based multimodal representation, and window length $w_{tr} = 8$, the worst-case computational cost is:

$$32 \cdot N_c^2 \cdot w_{tr} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This overhead is negligible relative to the cost of a single forward–backward pass through the model, so dissimilarity-based selection secures its diversity benefits at essentially no computational price.